

**GENERAL INSTRUCTIONS: Attempt any 60 Marks out of 70 Marks (Total Questions: 22)****SECTION - A : OBJECTIVE TYPE QUESTIONS****(1 Mark Each)**

- 1.(i). The number of significant figures in  $2.64 \times 10^{27}$  kg is: [1 Mark]
- (a) 3
  - (b) 4
  - (c) 24
  - (d) 1
- 1.(ii). The correct relation between Light year and metre is: [1 Mark]
- (a) 1 Light year =  $7.469 \times 10^{15}$  m
  - (b) 1 Light year = 4.2 m
  - (c) 1 Light year =  $9.467 \times 10^{15}$  m
  - (d) None of them
- 1.(iii). At what point in its trajectory does a projectile have minimum upward speed: [1 Mark]
- (a) At the start of the Trajectory
  - (b) At the end of the Trajectory
  - (c) At the midpoint of the Trajectory
  - (d) At the highest point in its Trajectory
- 1.(iv). Momentum is measured in: [1 Mark]
- (a)  $\text{kg s m}^{-1}$
  - (b)  $\text{kg m s}^{-2}$
  - (c)  $\text{kg m s}^{-1}$
  - (d) kg m
- 1.(v). A needle floats on the surface of water because of: [1 Mark]
- (a) Lighter weight
  - (b) Surface tension
  - (c) Adhesive force
  - (d) Viscosity
- 1.(vi). The meniscus of mercury in a glass tube is: [1 Mark]
- (a) Concave upward
  - (b) Convex upward
  - (c) Plane
  - (d) None of the above
- 1.(vii). The work done in an isothermal expansion of a gas depends upon: [1 Mark]
- (a) Temperature only
  - (b) Expansion ratio only
  - (c) Both of them
  - (d) None of them

- 1.(viii). On the basis of absolute zero as the scale of temperature, the ice melts at: [1 Mark]
- (a)  $100^{\circ}\text{C}$
  - (b)  $273.15\text{ K}$
  - (c)  $373.15\text{ K}$
  - (d) None of them
- 1.(ix). Two vessels A and B of the same size are at the same temperature, one of them holds  $1\text{ kg}$  of  $\text{H}_2$  gas and the other holds  $1\text{ kg}$  of  $\text{N}_2$  gas. Which of the vessels contains more molecules? [1 Mark]
- (a) A only
  - (b) B only
  - (c) Both (a) and (b)
  - (d) None of them
- 1.(x). The relation between pressure (P) and kinetic energy per unit volume (E) is: [1 Mark]
- (a)  $P = (2/3) E$
  - (b)  $P = (3/2) E$
  - (c)  $E = (1/2) \rho c^2$
  - (d) None of them

#### SECTION - B : VERY SHORT ANSWER QUESTIONS

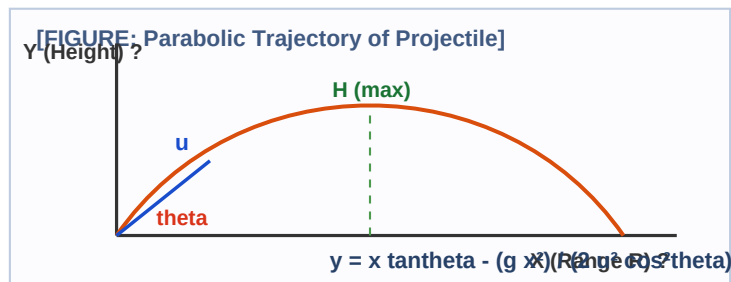
(2 Marks Each)

2. Define angular momentum. Give its SI unit and dimensional formula. [2 Marks]
3. Write the comparison of translational and rotational motion. [2 Marks]
4. State the physical significance of moment of inertia. [2 Marks]
5. Define surface energy. Write the relation between surface tension and surface energy. [2 Marks]
6. Is a reversible process really possible in practice? Discuss briefly. [2 Marks]
7. What are cyclic and non-cyclic thermodynamic processes? [2 Marks]
8. Write down the four basic assumptions of the Kinetic theory of gases. [2 Marks]
9. Write down the mathematical expression for the pressure of a gas in terms of density. [2 Marks]
10. What is meant by restoring force and force constant in elastic bodies? [2 Marks]

#### SECTION - C : SHORT ANSWER QUESTIONS

(3 Marks Each)

11. Write down the principal uses and limitations of dimensional analysis. [3 Marks]
12. Derive an equation for the trajectory of a projectile fired at an acute angle  $\theta$  with the horizontal (see diagram below): [3 Marks]



13. A cricket ball is rolled on ice with a velocity of 5.6 m/s and comes to rest after travelling 8 metres. Find the coefficient of friction. (Take  $g = 9.8 \text{ m/s}^2$ ). [3 Marks]
14. State Newton's first and second laws of motion. Also define one Newton of force. [3 Marks]
15. A body of mass 10 kg initially at rest is subjected to a force of 40 N. What is the kinetic energy acquired by the body at the end of 10 seconds? [3 Marks]
16. Distinguish clearly between elastic and inelastic collisions in one dimension. [3 Marks]
17. At what height above the Earth's surface will the acceleration due to gravity 'g' be the same as in a 100 km deep mine? [3 Marks]
18. Find an expression for the gravitational potential energy of a body situated at a distance 'r' from the centre of the Earth. [3 Marks]
19. At what distance from the mean position is kinetic energy equal to potential energy? Briefly explain in the case of Simple Harmonic Motion (S.H.M.). [3 Marks]

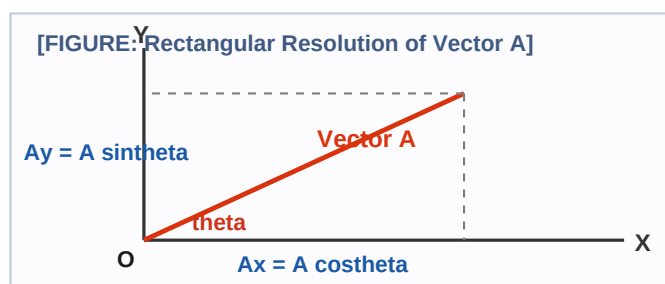
#### SECTION - D : LONG ANSWER QUESTIONS

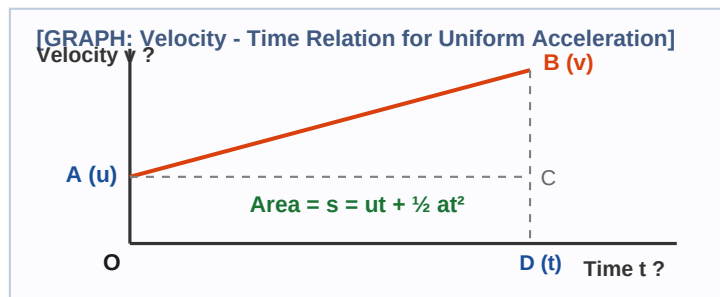
(5 Marks Each)

20. What do you mean by resolution of a vector? Find the rectangular components of a vector in two dimensions (see figure below): [5 Marks]

--- OR ---

From the velocity-time graph of uniform accelerated motion (see figure below), deduce the equations of motion in: (i) Velocity and time, and (ii) Position and time.





21. What do you mean by terminal velocity? Derive a mathematical expression for it using Stokes' Law.

[5 Marks]

--- OR ---

Explain all the three modes of transfer of heat (conduction, convection, radiation) with daily life examples.

22. Discuss the formation of stationary waves and harmonics in the case of open end organ pipes.

[5 Marks]

--- OR ---

Derive an expression for the instantaneous velocity and acceleration of a particle executing simple harmonic motion (S.H.M.).